

Fairness Beyond Demographics: Optimizing Performance Across Appearance-Based Hidden Cohorts in Medical Imaging

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Abstract. Medical image analysis models can exhibit performance disparities across patient subgroups, threatening clinical safety and fairness. Existing methods typically address this issue by optimizing accuracy and fairness metrics for visible demographic attributes (e.g., sex or age) considered in isolation. This strategy not only overlooks potentially more informative latent stratifications, which may reveal deeper sources of model error and inequity, but also fails to scale when multiple demographic attributes are considered simultaneously due to the resulting sparsity of training data within each subgroup. We deal with these issues by introducing the label-free hidden-cohort fairness (LHCF) training paradigm that instead of maximizing fairness over visible demographic attributes, it optimizes fairness across latent subpopulations discovered from image appearance. By clustering images into K appearance-based cohorts and applying fairness optimization over them, LHCF uncovers underlying sources of model error and avoids the combinatorial sparsity of multi-demographic attributes, reducing disparities across both single and multiple demographic attributes. We demonstrate on our proposed fairness benchmark, HIDFairBench, that LHCF provides state-of-the-art fairness results on single and multiple demographic attributes, despite never using demographic labels for training. Our results position hidden-cohort fairness as a practical, scalable, and robust alternative to demographic-based fairness optimization for trustworthy medical image analysis. Code will be available upon paper acceptance.

Keywords: Hidden Cohort Fairness · Demographic-Label-Free Fairness · Trustworthy AI in Healthcare.

1 Introduction

Medical image analysis (MIA) systems often exhibit performance disparities across visible demographic attributes (e.g., sex, age, ethnicity) [1, 6, 12, 18, 24, 32], raising concerns about clinical safety and fairness [6, 32]. Although these attributes are assumed to be the main drivers of such gaps, recent studies have shown that they may align only weakly with latent factors that may better explain model behavior [1, 24, 31]. Indeed, performance disparities can be much

larger between hidden patient cohorts (formed by visually coherent but unlabeled subsets of the data) than across demographic groups, creating low-performing subpopulations often referred to as underperforming slices or blind spots [10, 25, 27, 39]. This challenge becomes more severe when multiple attributes are considered jointly, as the number of intersectional subgroups expands while available samples per subgroup shrink [2]. These issues reflect the hidden stratification phenomenon, in which coarse labels fail to capture meaningful clinical variation [24, 40], causing models to perform poorly on underrepresented cohorts [8, 28, 29, 33] and ultimately increasing the risk of disproportionate patient harm [15].

Despite growing evidence that hidden patient cohorts represent a major driver of model failures, fairness methods in medical imaging still focus primarily on visible demographic attributes [19, 20, 22, 34, 35], while work on hidden cohorts has mostly diagnosed, instead of mitigating, the reported disparities [1, 5, 7, 25]. Demographic-focused fairness approaches also scale poorly as intersectional groups increase and per-group sample sizes collapse [2]. Taken together, these limitations point to latent cohorts, with sufficient data and richer structure, as a more reliable basis for fairness optimization than visible demographic groups. To illustrate this paradigm shift, we compare two approaches: (i) classic fairness training that optimizes across visible demographic attributes; and (ii) label-free hidden-cohort fairness (LHCF), which optimizes fairness across clusters formed with appearance-based embeddings. Fig. 1b shows that LHCF yields the best classification and fairness results not only for isolated demographic groups but also for multi-attribute intersections.

In this paper, we propose LHCF (Fig. 1a), which optimizes fairness over appearance-based latent cohorts without demographic labels and is compatible with existing fairness approaches. We also propose the HIDDEN-DEMOGRAPHIC FAIRNESS BENCHMARK (HIDFairBench), a new benchmark to stress-test fairness methods. This paper has the following contributions:

- The new LHCF training that clusters images into K hidden clusters and applies fairness optimization to balance accuracy and fairness across these clusters. This approach avoids the combinatorial sparsity of multi-attribute demographic groups and reduces performance disparities not only for the hidden clusters but also across single and multiple demographic attributes.
- The new HIDFairBench, a benchmark that evaluates hidden-cohort quality and tests fairness methods under increasingly complex multi-demographic partitions, exposing the scalability limits of fairness approaches.

Our experiments show that LHCF achieves state-of-the-art accuracy and fairness across single and multiple demographic attributes from HIDFairBench, despite never using demographic labels for training, establishing LHCF as a scalable and robust alternative to demographic-based fairness optimization in MIA.

2 Method

In this section, we present the LHCF training and the HIDFairBench benchmark.

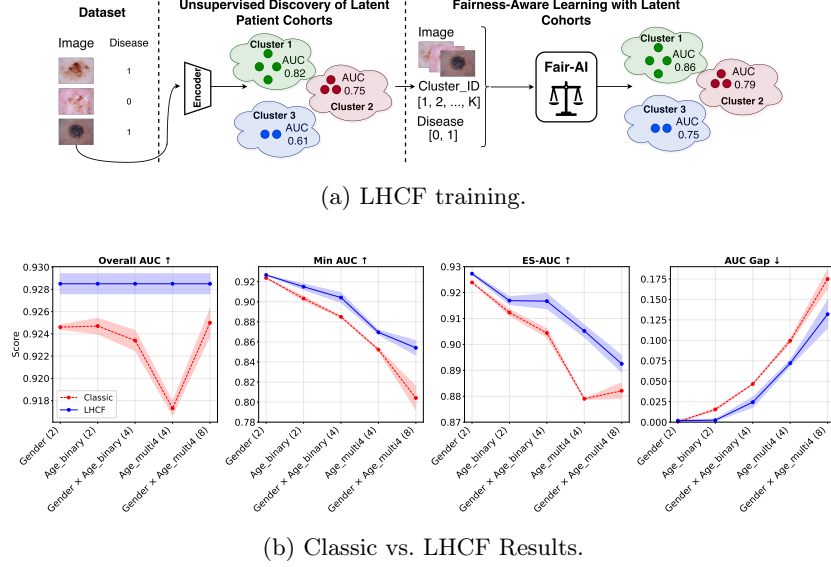


Fig. 1: (a) LHCF training. (b) Overall AUC (left) and fairness metrics (Min AUC, ES-AUC, AUC Gap; second-to-fourth plots) across visible cohort complexity for Classic demographic fairness training (i) and LHCF training (ii) of FairDi [22] (using ResNet18 backbone) on HAM10000 [36]. Visible-cohort complexity increases from Gender (2 groups) to Age_binary (2 groups), Age_multi4 (4 groups), and intersectional cohorts Gender×Age_binary (4 groups) and Gender×Age_multi4 (8 groups) – Sec. 2.2 explains this experimental setup.

2.1 LHCF Training

The **LHCF training** has two steps (Fig. 1a): **(i)** unsupervised discovery of latent cohorts from image embeddings, and **(ii)** fairness-aware training, where the discovered cohort identities are treated as sensitive attributes.

Step (i) assumes the availability of a labeled dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, containing medical images $x \in \mathcal{X}$ and diagnostic labels $y \in \mathcal{Y} \subseteq \Delta^{|\mathcal{Y}|-1}$, but without demographic annotations. Our goal is to learn the model $f_\theta = d_{\theta_d} \circ e_{\theta_e}$, where $\theta = (\theta_e, \theta_d)$, $e_{\theta_e} : \mathcal{X} \rightarrow \mathcal{Z}$ represents the encoder, and $d_{\theta_d} : \mathcal{Z} \rightarrow \mathcal{Y}$ denotes the decoder. We assume that the model $f_\theta(\cdot)$ has been pre-trained either with self-supervised learning [17] or with supervised learning using $\mathcal{D}$. In this Step (i), the method first finds the hidden cohorts via Gaussian Mixture Model (GMM) [9], where the number of clusters is estimated with the Bayesian Information Criterion (BIC) [30]. This clustering happens with the image embeddings in $\mathcal{Z} = \{z_i | z_i = e_{\theta_e}(x_i), (x_i, y_i) \in \mathcal{D}\}$. For $K \in \mathbb{N}$, a Gaussian mixture is defined by $p(z | \Phi_K) = \sum_{k=1}^K \pi_k \mathcal{N}(z | \mu_k, \Sigma_k)$, $\Phi_K = \{\pi_k, \mu_k, \Sigma_k\}_{k=1}^K$, with log-likelihood $\ell(\Phi_K; \mathcal{Z}) = \sum_{i=1}^N \log \sum_{k=1}^K \pi_k \mathcal{N}(z_i | \mu_k, \Sigma_k)$. We estimate Φ_K^* by Expectation Maximization (EM) [9], yielding the responsibility $\gamma_{ik} = p(k |$

$z_i, \Phi_K^*) = \frac{\pi_k \mathcal{N}(z_i | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(z_i | \mu_j, \Sigma_j)}$. The number of cohorts can be estimated with

$$K^* \in \operatorname{argmin}_{K \in \{1, \dots, N\}} \text{BIC}(K) = \operatorname{argmin}_{K \in \{1, \dots, N\}} p_K \log N - 2 \ell(\Phi_K^*; Z), \quad (1)$$

where $p_K = (K - 1) + Kd + K \frac{d(d+1)}{2}$ for full covariances. The hidden cohort dataset to be used in subsequent fairness optimization is then defined by $\hat{\mathcal{D}} = \{(x_i, y_i, c_i) | (x_i, y_i) \in \mathcal{D}, c_i = \arg \max_{k \in \{1, \dots, K^*\}} \gamma_{ik}\}$, where c_i represents the hidden cohort label of the sample (x_i, y_i) computed with the responsibility γ_{ik} .

Step (ii) uses the hidden cohort dataset $\hat{\mathcal{D}}$ to optimize

$$\theta^* = \operatorname{argmin}_{\theta} \frac{1}{N} \sum_{i=1}^N \ell_{\text{class}}(y_i, f_{\theta}(x_i)) + \lambda \ell_{\text{fair}}(\hat{\mathcal{D}}, \theta), \quad (2)$$

where $\lambda \geq 0$ is a hyper-parameter, $\ell_{\text{class}}(\cdot)$ is a classification loss and $\ell_{\text{fair}}(\cdot)$ denotes a fairness loss that can be defined in different ways, such as:

$$\ell_{\text{fair}}^{\text{wst}}(\hat{\mathcal{D}}, \theta) = \max_k R_k(\theta) \quad \text{or} \quad \ell_{\text{fair}}^{\text{gap}}(\hat{\mathcal{D}}, \theta) = \max_k R_k(\theta) - \min_k R_k(\theta), \quad (3)$$

with $R_k(\theta) = \frac{1}{|C_k|} \sum_{(x_i, y_i) \in C_k} \ell_{\text{class}}(y_i, f_{\theta}(x_i))$ denoting classification loss of samples in the cluster C_k , $\ell_{\text{fair}}^{\text{wst}}(\cdot)$ minimizing the worst-cohort classification loss, and $\ell_{\text{fair}}^{\text{gap}}(\cdot)$ minimizing the gap between the best- and worst-cohort losses. LHCF is agnostic to the model and fairness loss, which together define a FairAI method (e.g., SWAD [4], FIS[20], FEBS [35], FairCLIP [19], FaMI [34], FairDi [22]).

Theoretical analysis We show that under mild clustering assumptions, the fairness loss $\ell_{\text{fair}}^{\text{wst}}(\hat{\mathcal{D}}, \theta)$ upper-bounds the loss of any visible cohort; thus, minimizing it yields fairness across all visible cohorts [29]. Specifically, we assume that the GMM with hard assignments partitions the data into K disjoint clusters $\{C_1, \dots, C_K\}$ with $C_k \cap C_{k'} = \emptyset$ for $k \neq k'$, and that each visible cohort G_t is a union of these clusters, i.e., $G_t = \bigcup_{k \in \mathcal{K}} C_k$, where $\mathcal{K} \subseteq \{1, \dots, K\}$.

Lemma 1. *The empirical risk of any visible cohort is upper-bounded by $\ell_{\text{fair}}^{\text{wst}}(\hat{\mathcal{D}}, \theta)$, meaning: $\mathbb{L}(G_t) = \frac{1}{|G_t|} \sum_{(x_i, y_i) \in G_t} \ell_{\text{class}}(y_i, f_{\theta}(x_i)) \leq \max_k R_k(\theta)$.*

Proof. Assuming $G_t = \bigcup_{k \in \mathcal{K}} C_k$, $\mathbb{I}((x_i, y_i) \in G_t) = \sum_{k \in \mathcal{K}} \mathbb{I}((x_i, y_i) \in C_k)$, where $\mathbb{I}(\cdot)$ is the indicator function, we re-write the risk on G_t as:

$$\begin{aligned} \mathbb{L}(G_t) &= \frac{1}{|G_t|} \sum_{(x_i, y_i)} \ell_{\text{class}}(y_i, f_{\theta}(x_i)) \mathbb{I}((x_i, y_i) \in G_t) \\ &= \frac{1}{|G_t|} \sum_{(x_i, y_i)} \sum_{k \in \mathcal{K}} \ell_{\text{class}}(y_i, f_{\theta}(x_i)) \mathbb{I}((x_i, y_i) \in C_k) \\ &= \frac{1}{|G_t|} \sum_{k \in \mathcal{K}} \sum_{(x_i, y_i) \in C_k} \ell_{\text{class}}(y_i, f_{\theta}(x_i)) \\ &= \frac{1}{|G_t|} \sum_{k \in \mathcal{K}} |C_k| R_k(\theta) \leq \sum_{j \in \mathcal{K}} |C_j|/|G_t| \max_k R_k(\theta) \leq \max_k R_k(\theta). \end{aligned} \quad (4)$$

□

2.2 HIDFairBench Benchmark

Datasets. The benchmark relies on three public medical imaging datasets: *Fitzpatrick17K* [14], *HAM10000* [36], and *CMMD* [3]. For HAM10000 (9,948 dermoscopic images), we follow prior work [21] and binarize diagnoses into

benign vs. malignant. Visible cohort partitions include *Gender*, *Age_binary* (≤ 60 , > 60), *Age_multi4* (≤ 39 , 40–59, 60–79, ≥ 80), and intersectional *Gender* $\times$ *Age_binary* and *Gender* $\times$ *Age_multi4* groups. For Fitzpatrick17K, we group lesions into benign (non-neoplastic + benign) vs. malignant, and use Fitzpatrick *skin types* (0–5) as visible demographic attributes. For both HAM10000 and Fitzpatrick17K, we use an 80%/10%/10% train/validation/test split. For CMMD (5,152 mammography images), we frame classification as non-cancer vs. cancer, use binarized *patient age* (≤ 55 , > 55) as the sensitive attribute, and rely on a 70%/10%/20% train/validation/test split.

Evaluation & Statistical Tests. The hidden-cohort quality is measured using the *Brier Score (BS)* [25], which captures calibration and identifies under- and overperforming cohorts, and *Average Purity (AP)* [1], which quantifies alignment between discovered cohorts and visible demographic attributes. For visible cohort evaluation, we report *AUC* and four fairness metrics: *AUC Gap*, *Worst-Case AUC* [41], *ES-AUC* [20], and *Performance-Scaled Disparity (PSD)*. Statistical significance uses the *Friedman test* [13, 41] with *Nemenyi* post-hoc comparisons [23] ($p < 0.05$), visualized via Critical Difference (CD) diagrams.

FairAI Methods & Backbones. The benchmark tests many *FairAI methods*, starting from the *ERM* [37] baseline, which optimizes average classification error without addressing fairness. *SWAD* [4] improves generalization by promoting convergence to flat minima. Among fairness approaches, *FIS* [20] reweights group losses to equalize performance, while *FEBS* [35] emphasizes hard examples via error-bound scaling. We also include *FairCLIP-OT*, an optimal-transport fairness regularizer adapted from FairCLIP [19], and *FaMI* [34], which reduces dependence between representations and sensitive attributes via mutual information. Furthermore, *FairDi* [22] learns a shared fair backbone and distills knowledge from cohort-specific experts into a unified student model. For the *backbone models*, we consider *ResNet18* [16], *CLIP* [11], *MedCLIP* [38], and *DINOv2* [26].

3 Experiments

3.1 HIDFairBench Results

Hidden-Cohort Quality. Using an ERM-trained ResNet18, BS on HAM10000 reveals a clear risk stratification across hidden cohorts (Table 1): low malignancy-rate cohorts (Clusters 0,2) show low BS, intermediate malignancy-rate cohorts (1,3,5) exhibit moderate BS, and high malignancy-rate cohorts (4,6) attain the highest BS, indicating reduced reliability on clinically severe cases. These trends mirror underlying population structure, with cohorts dominated by older patients showing higher malignancy prevalence, consistent with known age-related risk patterns in dermatology. Across CMMD and Fitzpatrick17K, BS similarly tracks malignancy severity, with low malignancy-rate cohorts obtaining low calibration error and high malignancy-rate clusters showing higher BS. Despite this clear risk stratification, hidden cohorts show weak alignment with available demographic metadata: on HAM10000, AP is 0.4919 ± 0.0183 (gender) and

Table 1: AUC and BS across hidden clusters (Clt. ID) on HAM10000 (ERM, ResNet18), with cohort proportions (%) of malignant/benign, male/female, and age ≤ 60 or > 60 years old.

Clt.ID	AUC	BS	(Mal)(Ben)	(Male)(Female)	(≤ 60)(> 60)
0	–	5.3E-5	(0.00)(100.00)	(48.66)(51.34)	(78.61)(21.39)
1	0.6179	0.1506	(18.55)(81.45)	(39.18)(60.82)	(70.10)(29.90)
2	0.8247	0.0241	(2.63)(97.37)	(46.24)(53.76)	(74.44)(25.56)
3	0.7507	0.1759	(27.27)(72.73)	(45.45)(54.55)	(54.54)(45.46)
4	0.6113	0.2255	(36.36)(63.64)	(63.64)(36.36)	(35.06)(64.94)
5	0.7557	0.1090	(13.65)(86.35)	(59.03)(40.97)	(56.38)(43.62)
6	0.7299	0.1956	(66.39)(33.61)	(69.75)(30.25)	(23.53)(76.47)

Table 2: Classic vs. LHCF training results averaged over HAM10000 (*Gender*, *Age_binary*, *Age_multi4*, *Gender* $\times$ *Age_binary*, *Gender* $\times$ *Age_multi4*), Fitzpatrick17K (*Skin_Type*), and CMMD (*Age_binary*). Cell colors denote **Better** / **Worse** / **Same** between LHCF and Classic, and **Bold**=best result.

	Measures and Ranks	ERM	SWAD	FIS	FaMI	FEBS	FairCLIP-OT	FairDi
Classic	Avg. Overall AUC Score $\uparrow$	0.8775	0.8867	0.8806	0.8582	0.8731	0.8692	0.9014
	Avg. Min. AUC Score $\uparrow$	0.8213	0.8340	0.8212	0.8113	0.8246	0.8083	0.8511
	Avg. ES-AUC Score $\uparrow$	0.8447	0.8578	0.8477	0.8304	0.8444	0.8374	0.8744
	Avg. Gap AUC Score $\downarrow$	0.1053	0.0905	0.1019	0.0953	0.0940	0.1012	0.0824
	Avg. Mean PSD Score $\downarrow$	0.0478	0.0406	0.0487	0.0448	0.0444	0.0473	0.0354
	Avg. Max PSD Score $\downarrow$	0.1220	0.1042	0.1186	0.1134	0.1104	0.1192	0.0915
LHCF	Avg. Overall AUC Score $\uparrow$	0.8775	0.8889	0.8804	0.8668	0.8827	0.8703	0.9050
	Avg. Min. AUC Score $\uparrow$	0.8213	0.8418	0.8266	0.8106	0.8369	0.8127	0.8686
	Avg. ES-AUC Score $\uparrow$	0.8447	0.8617	0.8486	0.8339	0.8519	0.8399	0.8817
	Avg. Gap AUC Score $\downarrow$	0.1053	0.0843	0.0967	0.1061	0.0929	0.0963	0.0666
	Avg. Mean PSD Score $\downarrow$	0.0478	0.0386	0.0461	0.0498	0.0450	0.0451	0.0312
	Avg. Max PSD Score $\downarrow$	0.1220	0.0976	0.1124	0.1245	0.1080	0.1132	0.0762

0.2234 $\pm$ 0.0065 (age); on Fitzpatrick17K, AP for skin type is 0.0998 $\pm$ 0.0161; and on CMMD, AP for age is 0.3517 $\pm$ 0.0067. Such APs < 0.5 confirm that hidden cohorts primarily reflect visual structure rather than demographic attributes.

Visible Cohort Evaluation. Table 2 compares visible-demographic supervised (*Classic*) and *LHCF* training across all datasets and demographic attributes. LHCF improves AUC and fairness, especially for intersectional cohorts, without using any demographic metadata (see Fig. 1b). Despite weak alignment between hidden and demographic groups (AP < 0.5), the fairness learned from appearance-based cohorts generalizes well to visible attributes, with FairDi+LHCF delivering the strongest overall results. The CD diagrams in Fig. 2 confirm the superiority of LHCF, and in particular the FairDi+LHCF as the top-ranked approach across nearly all evaluation metrics. An important question is why LHCF, trained on hidden cohorts, outperforms classical methods trained on visible cohorts. The intuition, based on Lemma 1, is that instead of minimizing the average loss over overlapping visible cohorts, which can overemphasize intersectional groups, LHCF minimizes the loss over hidden cohorts, thereby reducing extreme disparities and producing more balanced performance.

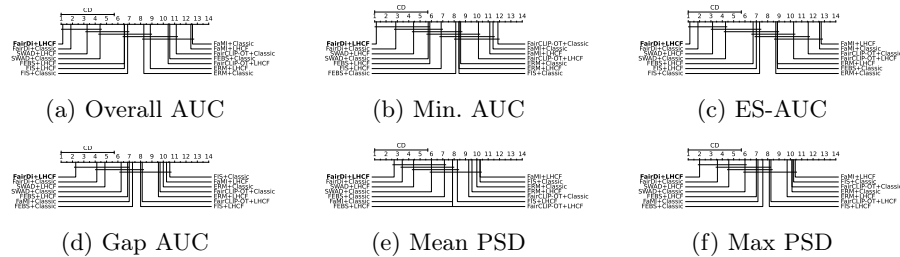


Fig. 2: CD diagrams comparing Classic and LHCf learning across all datasets and demographic cohorts (Please zoom in to view the figures clearly).

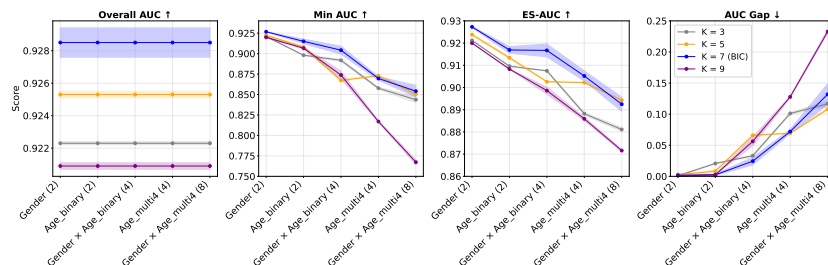


Fig. 3: Sensitivity analysis of LHCf (FairDi+ResNet18 backbone) with respect to the number of hidden cohorts (K) on the HAM10000 dataset.

3.2 Ablation Studies

Number of Hidden Cohorts (K). Fig. 3 evaluates the performance of LHCF as a function of the number of cohorts. The BIC choice (i.e., $K=7$) produces the best results for AUC and fairness measures for the majority of visible cohort partitions, whereas larger or smaller values of $K \in \{3, 5, 9\}$ tend to degrade performance. This supports BIC as an effective criterion for selecting K .

Embedding Backbone. Fig. 4 summarizes how representation backbones affect LHCF. ResNet18 and MedCLIP achieve the best Overall AUC. Fairness metrics separate more clearly: for Min. AUC and AUC Gap, ResNet18 performs best on simpler intersectional settings; for ES-AUC, ResNet18 is consistently strongest across all cohort complexities, with MedCLIP a solid second. CLIP and DINOv2 show larger drops in worst-case performance and higher disparity as cohort complexity increases. Overall, backbones inducing *moderate* hidden-cohort granularity, e.g., ResNet18 ($K=7$) and MedCLIP ($K=6$) selected by BIC, yield better accuracy and fairness than CLIP ($K=10$) and DINOv2 ($K=5$).

Demographic-Aware Clustering (DAC) vs. LHCF. As an intermediate alternative between *Classic* and *LHCF*, DAC clusters samples using embeddings augmented with demographic attributes and then optimizes fairness over these clusters. Fig. 5 shows that LHCF outperforms DAC on nearly all measures and across all cohort complexities. These results indicate that appearance driven.

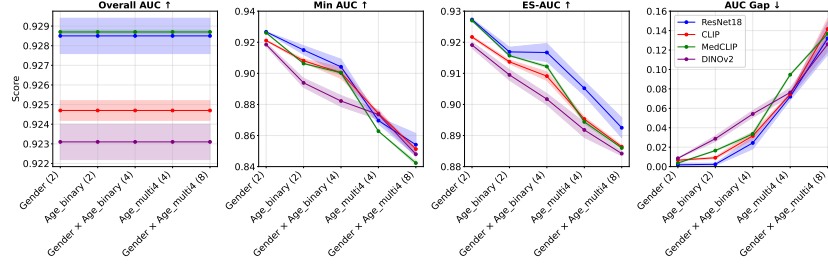


Fig. 4: Sensitivity analysis of LHCF (FairDi) with respect to the backbone (ResNet18, CLIP [11], MedCLIP [38], and DINOv2 [26]) on HAM10000.

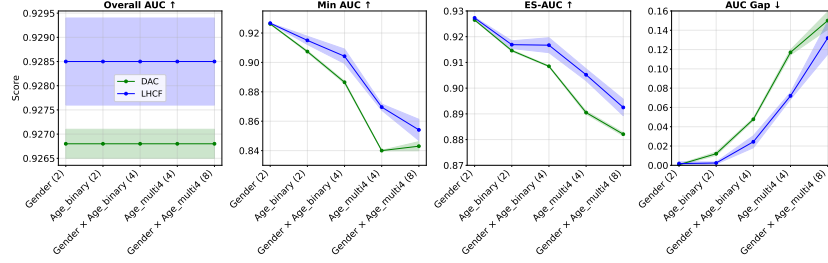


Fig. 5: DAC vs. LHCF (FairDi+ResNet18 backbone) on HAM10000 dataset.

label-free fairness optimization offers greater robustness to visible-cohort complexity than DAC while eliminating any dependence on demographic labels.

4 Conclusion

We introduced LHCF, which optimizes fairness over appearance-based latent cohorts without demographic supervision, and proposed *HIDFairBench* to evaluate fairness under single and multiple demographic attributes. Across three datasets, hidden cohorts exhibit clinical risk stratification (BS rises with malignancy) but weak demographic alignment ($AP < 0.5$), indicating they capture meaningful visual structure. Relative to the demographic-supervised Classic FairAI methods, LHCF improves fairness and accuracy on visible groups without using demographic labels, with solid gains for intersectional cohorts. LHCF is robust to design choices: a BIC-selected moderate number of cohorts ($K=7$) yields the best accuracy-fairness tradeoff, and backbones with similar cohort granularity (ResNet18, MedCLIP) provide the best performance. Compared with DAC, LHCF delivers better accuracy and fairness. In future work, we will improve the reproducibility of LHCF by stabilizing the unsupervised clustering stage and extend Lemma 1 to accommodate more fairness losses; nonetheless, the adoption of data-driven latent stratification offers a more generalizable and deployable pathway toward equitable and trustworthy medical imaging analysis.

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